

Input data documentation

1. 100m x 100m grid-based emission

File: SubIEGrid_new_SummarizeWithin_TableToExcel_1.csv

Description:

This file contains grid-level emissions aggregated from SOCAL BEAM link outputs using spatial overlay (ArcGIS *Summarize Within*). Each row represents a 100 m × 100 m grid cell.

Variable	Description	Unit
gridID	Unique grid identifier	–
point_x, point_y	Grid centroid coordinates (longitude, latitude)	WGS degrees
sum_totalbylink0316_csv_PM25_truck	Daily PM2.5 emissions from trucks	grams/day
sum_totalbylink0316_csv_PM25_car	Daily PM2.5 emissions from passenger cars	grams/day
sum_totalbylink0316_csv_NOx_truck	Daily NOx emissions (truck)	grams/day
sum_totalbylink0316_csv_NOx_car	Daily NOx emissions (car)	grams/day
sum_totalbylink0316_csv_ROG_*	Reactive organic gases	grams/day
sum_totalbylink0316_csv_SOx_*	Sulfur oxides	grams/day
sum_totalbylink0316_csv_NH3_*	Ammonia	grams/day
pop (<i>optional</i>)	Population in grid cell	persons

This input file does not include UTM coordinates; therefore, it must be projected to UTM Zone 10N (EPSG:26910) for the San Francisco Bay Area.

Processing Notes

- Emissions are converted to annual short tons using:

$$E_{annual} = \frac{E_{daily} \times 330}{907184.74}$$

Where the 330 days are taken from ISRM calculation. 907184.74 means annual short ton at emission rate of 0.1 gram/m²/second.

- Truck and passenger car emissions are summed at grid level
- Missing values are set to 0

2. AERMOD dispersion pattern library

File: full_aggregate_data_wgs.csv

Description:

This file contains post-processed AERMOD simulation outputs, representing dispersion response functions (i.e., concentration per unit emission).

Each row corresponds to a receptor point relative to a source location.

Variable	Description	Unit
DataSet_ID	AERMOD simulation site (e.g., LIVERMORE_2015)	–
Grid_X, Grid_Y	Relative coordinates from source center	UTM 10N meters
Distance	Distance from source	Kilo meters
Concentration	Modeled concentration, needs unit converting	g/m ² /s
Height	Emission release height	meters
Urban_Rural	Population/urban classification	persons
Emissions	Emission type (e.g., CITYSTREET, FREEWAY)	–

Unit Conversion

The concentration is converted to:

$$\mu\text{g}/\text{m}^3 \text{ per annual short ton}$$

using:

$$C' = \frac{C}{(wd_{\text{annual}}/907184.74)}$$

3. ISRM concentration product

Files: SubIEGrid_new_with__ISRM.csv

Description:

This file contains ISRM-based PM2.5 concentration components at 100m x 100m grid level. It is used to integrate secondary PM formation with AERMOD-based primary PM estimates.

Variable	Description	Unit
gid	Grid ID	–
SOA	Secondary organic aerosol	µg/m ³
pNO3	Nitrate PM	µg/m ³
pNH4	Ammonium PM	µg/m ³
pSO4	Sulfate PM	µg/m ³
PrimaryPM25	Primary PM2.5	µg/m ³
TotalPM25	Total PM2.5	µg/m ³

Model Integration Logic

Total PM2.5

$$PM2.5_{total} = PM2.5_{AERMOD\ primary} + PM2.5_{secondary}^{ISRM}$$